

HNSS12 · Oral Structures — Histology Chart

The Oral Cavity & Pharynx				
Functions gateway to the GI tract · articulation & resonance of speech · secondary respiration + defense		Lining mucosa non-keratinized: cheeks, inner lips, floor of mouth, ventral tongue, soft palate		Masticatory mucosa keratinized: gingiva, hard palate, dorsal tongue
STRUCTURE	LOCATION / EXTENT	HISTOLOGY (SLIDE ID)	FUNCTION	KEY FACT / CLINICAL
Oral mucosa <i>the wet lining</i> ALL SURFACES	Lines the vestibule and oral cavity proper	Stratified squamous , mostly non-keratinized , with no zonation (no strata). Keratinized only on masticatory (friction) surfaces: gingiva, hard palate, dorsal tongue	Protection; stays chronically moist	Highly permeable , especially the floor of the mouth and ventral tongue → sublingual nitroglycerin (skips first-pass metabolism)
Lip: external surface <i>the skin side</i> LIP	Outer face of the lip	Skin : keratinized stratified squamous with hair + sweat glands	Protection	—
Lip: vermillion zone <i>the red margin</i> LIP	Red border between skin and mucosa	Transition from keratinized → non-keratinized epithelium; abundant dermal capillaries ; no hair or sweat glands	Red lip margin	Red because thin epithelium sits over many capillaries; no glands → dries out (licking)
Lip / cheek: internal <i>the inner lining</i> LIP + CHEEK	Inner lips and cheeks	Typical oral mucosa (non-keratinized) tethered to the underlying muscle at intervals	Allows lip and cheek movement	Many small folds instead of one big fold that could be bitten
Hard palate <i>the bony roof</i> PALATE	Anterior 2/3 of the palate	Keratinized mucosa tightly bound to the periosteum of bone; numerous minor mucous glands	Separates the nasal from the oral cavity ; chewing surface	Masticatory mucosa
Soft palate <i>the moving curtain</i> PALATE	Posterior 1/3 ; ends in the uvula	Non-keratinized mucosa over muscle; many taste buds	Separates the nasopharynx from the oropharynx ; rises to seal the nasopharynx when swallowing	CN X lesion → uvula deviates away from the lesion STEP 1
Gingiva <i>the gums</i> TEETH	Around the necks of the teeth	Keratinized oral epithelium firmly bound to the periosteum of the alveolar processes. Gingival sulcus (space between free gingiva and crown) is lined by non-keratinized epithelium	Seals around the tooth	Plaque builds up in the sulcus. Gingival hyperplasia: phenytoin , CCBs, cyclosporine STEP 1

The three oral spaces

SPACE	WHERE
Vestibule	Between lips/cheeks and teeth
Oral cavity proper	Inside the dental arches
Pharynx	Behind, toward the esophagus and larynx

The pharynx: epithelium follows function

REGION	CARRIES	EPITHELIUM	KEY FACT
Nasopharynx	Air : nasal cavity → larynx + trachea	Pseudostratified ciliated columnar with goblet cells (respiratory)	Pharyngeal tonsil (adenoids). Nasopharyngeal carcinoma = EBV STEP 1
Oropharynx	Food : oral cavity → esophagus	Non-keratinized stratified squamous (abrasion-resistant)	Palatine tonsils. HPV-16 SCC (tonsils, base of tongue) STEP 1

★ **USMLE high-yield**: Oral mucosa is **non-keratinized** except where food rubs (**gingiva, hard palate, dorsal tongue**). **Vermilion** = capillary-rich transition zone with **no hair or sweat glands**. **Hard palate** = keratinized mucosa on bone; **soft palate** = taste buds + uvula. **Nasopharynx** = respiratory epithelium (air); **oropharynx** = non-keratinized stratified squamous (food). Floor of mouth is the most permeable site → sublingual drugs.

Teeth: Tissues, Organization & Development

32 permanent

8 per quadrant; preceded by 20 deciduous

Crown · neck · root

crown = enamel; root = cementum; they meet at the neck

No deciduous molars

permanent molars have no baby-tooth precursor

STRUCTURE	LOCATION / EXTENT	HISTOLOGY (SLIDE ID)	FUNCTION	KEY FACT / CLINICAL
Enamel <i>the hardest tissue</i> CROWN	Covers the crown ; meets cementum at the neck	95% hydroxyapatite — hardest, most calcium-rich tissue in the body; long, tightly packed enamel rods span its full thickness	Biting / grinding surface	Made by ameloblasts , which degenerate at eruption → enamel cannot repair . Fluoride → acid-resistant fluorapatite STEP 1
Dentin <i>the bulk of the tooth</i> CROWN + ROOT	Under enamel (crown) and cementum (root); surrounds the pulp	Hard but resilient, 70% mineral . Odontoblasts line its inner (pulp) surface , make predentin (type I collagen) that mineralizes; their processes run in dentinal tubules	Main body of the tooth; supports enamel	Odontoblasts are long-lived → dentin forms for life and the pulp cavity shrinks. Osteogenesis imperfecta → dentinogenesis imperfecta STEP 1
Pulp <i>the living core</i> PULP CAVITY	Pulp cavity + root canal ; enters at the apical foramen	Gelatinous connective tissue, highly vascular and innervated; all afferents are pain fibers (some enter dentinal tubules)	Nourishes odontoblasts; sensation	Rigid chamber → inflammatory swelling compresses nerves → toothache (pulpitis)
Cementum <i>the root's glue</i> ROOT	Covers the dentin of the root	Bone-like calcified matrix made by cementocytes (like osteocytes)	Anchors Sharpey's fibers from the periodontal ligament	Labile : remodels in response to pressure (orthodontics)
Periodontal ligament <i>the shock absorber</i> ROOT	Between cementum and alveolar bone	Dense connective tissue = the periosteum of the alveolar bone ; sends type I collagen (Sharpey's) fibers into cementum	Secures the tooth in its socket; shock absorber ; allows slight movement	Periodontal disease = main cause of adult tooth loss STEP 1
Alveolar bone <i>the socket</i> JAW	Alveolar processes of the maxilla + mandible	Woven bone projecting between teeth to form the sockets	Holds the roots	—

Tooth layers, superficial → deep

Ameloblasts (if present) → Enamel → Dentin → Predentin → Odontoblasts → Pulp

Alveolar bone → PDL (Sharpey's fibers) → Cementum → Dentin → Pulp

Development: starts **weeks 6–7** of gestation; **bud** → **cap** → **bell**; **crown forms before root**, root forms with **eruption**.

Dentigerous cyst

Most common **non-inflammatory odontogenic cyst**. Around the **crown of an unerupted / impacted tooth** (lower 3rd molar, upper canine), attached at the neck; fluid between the follicle and reduced enamel epithelium. Can transform into **ameloblastoma**, **SCC** or **mucoepidermoid carcinoma**.

Tooth types

TOOTH	NUMBER (PERMANENT)	JOB	ROOTS
Incisors	8	Cutting	1
Canines	4	Grasping	1
Premolars	8	Grinding	2
Molars	12	Grinding	2–3

What each embryonic part makes

EMBRYONIC PART	ORIGIN	MAKES
Enamel organ	Ectoderm (down-growth of oral epithelium)	Ameloblasts → enamel
Dental papilla	Mesoderm per lecture (neural crest ectomesenchyme STEP 1)	Odontoblasts → dentin + pulp
Dental sac	Mesenchyme around papilla + enamel organ	Cementoblasts + periodontal ligament

★ **USMLE high-yield: Enamel** = 95% hydroxyapatite, made by **ameloblasts** (ectoderm) that die at eruption → **no repair**. **Dentin** = odontoblasts on the pulp side, forms for life. **Cementum + PDL (Sharpey's fibers)** anchor the root in woven alveolar bone. Hardness: enamel 95% > dentin 70% > cementum ≈ bone. Tetracycline → discolored teeth; fluoride excess → fluorosis.

The Tongue, Papillae & Taste Buds

STRUCTURE	LOCATION / EXTENT	HISTOLOGY (SLIDE ID)	FUNCTION	KEY FACT / CLINICAL
Tongue body <i>the muscular core</i> WHOLE TONGUE	Anterior 2/3 (oral) + posterior 1/3 (pharyngeal), split by the V-shaped sulcus terminalis	Interlacing skeletal muscle in 3 planes EXAM EMPHASIS covered by mucosa. Dorsal : thick keratinizing epithelium with papillae. Ventral : smooth, thin, permeable	Articulation, mastication, taste, swallowing, suckling	Triplanar muscle = slide ID for tongue
Filiform papillae <i>the friction spikes</i> ANTERIOR 2/3	Everywhere on the dorsum; most numerous	Conical, keratinized	Friction for moving food	No taste buds
Fungiform papillae <i>the scattered mushrooms</i> ANTERIOR 2/3	Scattered among filiform, mostly near the tip	Mushroom-shaped , thin epithelium on top	Taste	A few taste buds on the upper surface
Circumvallate papillae <i>the moated giants</i> ANTERIOR 2/3	10–12 in a V just in front of the sulcus terminalis	Largest papillae, each surrounded by a moat ; non-keratinized. Von Ebner glands (serous) lie beneath	Most taste : >50% of all taste buds, in the moat walls	Von Ebner glands flush the moat (cleansing); the only serous minor salivary glands
Lingual tonsils <i>the cobblestones</i> POSTERIOR 1/3	Pharyngeal (posterior 1/3) surface	Sub-epithelial lymphoid nodules ; each has one deep crypt	Immune defense (part of Waldeyer's ring)	Minor salivary glands drain into the crypts (unlike other tonsils)

Taste bud: ~50 spindle cells in an intraepithelial ellipsoid

CELL	WHAT IT IS	KEY FACT
Sensory (gustatory)	Elongated cells (~half the bud) with microvilli projecting into the taste pore ; synapse on gustatory nerve fibers	~10-day life span (high turnover)
Supporting	Support cells	—
Basal	Stem cells	Replace sensory cells

Taste pore = apical ends of sensory cells joined by **tight junctions**. Five tastes: **sweet, sour, salty, bitter, umami**. Taste buds also on the **soft palate** (uvula), pharyngeal wall, epiglottis, larynx; a few in lips, cheeks, upper esophagus. **Number falls with age**.

Tongue innervation **STEP 1**

REGION	TASTE	TOUCH
Anterior 2/3	VII (chorda tympani)	V3 (lingual)
Posterior 1/3	IX	IX
Epiglottis region	X	X
Muscles	XII (palatoglossus = X)	—

All taste → solitary nucleus → VPM thalamus. Chemo/radiation kill fast-turnover taste cells → taste loss. Zinc deficiency → dysgeusia.

★ **USMLE high-yield**: Tongue = **tripplanar skeletal muscle**. **Filiform** = most numerous, keratinized, **no taste buds**. **Fungiform** = a few. **Circumvallate** = >half of taste buds + **von Ebner (serous) glands** in a V before the sulcus terminalis. **Lingual tonsils**: one crypt each, minor salivary glands drain into the crypts. Sensory taste cells last ~10 days, replaced by basal stem cells.

Salivary Glands

Saliva

~95% water + proteins (amylase, mucus, IgA, lysozyme, lactoferrin) + K^+ , Ca^{2+} , HCO_3^- ; ~1 L/day

Parasympathetic

copious,
watery
saliva

Sympathetic

scant,
thick
, viscous saliva ("dry mouth")

The three major glands: compare and identify on a slide (flagged in review)

FEATURE	PAROTID	SUBMANDIBULAR	SUBLINGUAL
Share of saliva	~25%	~70%	~5%
Location / size	Largest ; next to the ear	Along the body of the mandible	Smallest ; floor of the mouth
Classification	Compound acinar (alveolar)	Compound tubuloalveolar	Compound tubuloalveolar
Secretory cells	All serous (dark, granular)	Mixed, mainly serous	Mixed, mainly mucous (pale)
Serous demilunes	None	Frequent — diagnostic	Some
Capsule	Heavy	Less prominent	None
Intercalated ducts	Abundant	Less prominent	Absent
Striated ducts	Abundant	Abundant	Rare
Interstitial fat	Abundant	Little / none	None
Duct	Stenson's → inner cheek opposite the upper 2nd molar	Wharton's → floor of the mouth	Bartholin's / many ducts, behind Wharton's
Extra	Many IgA plasma cells ; secretes water, electrolytes, amylase	Saliva fairly rich in mucin; most stones STEP 1	Saliva rich in mucin

Cell types

CELL	LOOK ON H&E	SECRETES / DOES
Serous	Dark, granular (zymogen granules); round central nucleus	Water, electrolytes, amylase , IgA, lactoferrin
Mucous	Pale, foamy ; flattened basal nucleus	Mucin (lubrication)
Myoepithelial	Flat cells wrapped around acini	Squeeze saliva out
Plasma cell	In the stroma	IgA

Major vs minor glands

	MAJOR	MINOR
Examples	Parotid, submandibular, sublingual	Lips, cheeks, palate, tongue (submucosa)
Capsule	Encapsulated, grossly visible	Not encapsulated, small
Secretion	On demand	Continuous ; short ducts; mostly mucous (von Ebner = serous)

Two-stage secretion

Acinus: primary secretion (myoepithelial help) → Intercalated duct → **Striated duct**: modifies ions →

Excretory duct

Striated ducts = columnar cells with **basal folds packed with mitochondria** → active electrolyte transport. Final saliva is **hypotonic** (Na^+/Cl^- reabsorbed, K^+/HCO_3^- secreted) **STEP 1**

Slide ID in one line each

Parotid: all dark serous acini + **fat cells** + many ducts.

Submandibular: dark serous + pale mucous mix with **serous demilunes**.

Sublingual: mostly **pale mucous** acini, few serous, no capsule.

★ **USMLE high-yield**: **Parotid** = pure serous, 25%, Stenson's (upper 2nd molar). **Submandibular** = mixed (mostly serous, demilunes), **70%**, Wharton's. **Sublingual** = mixed (mostly mucous), 5%, Bartholin's. Parasympathetics: parotid via **IX** → **otic ganglion**; submandibular + sublingual via **VII** → **submandibular ganglion**. Sialolithiasis → submandibular. Mumps → parotitis. Pleomorphic adenoma = most common salivary tumor.